intermediate trainer jets. The final stage of training happens aboard the British-made Hawk Advanced Jet trainer. After this, they are deemed sufficiently capable to handle frontline fighter jets of the IAF like Sukhois and Rafales.

The journey of the HTT-40, say its makers, has been full of challenges and roadblocks. HAL, India's only state-owned aerospace company, took years to overcome the obstacles and convince the ministry of defence (MoD) and the IAF of the HTT-40's potential. HAL officials say it is only the Modi government's determined Make in India push in defence hardware that salvaged the project, after years of it being doubted by the IAF, its end user.

The HTT-40 is a twin-seater, fully aerobatic turboprop (i.e., propeller-driven) aircraft with an air-conditioned cockpit and modern avionics. It is capable of being hot-refuelled (while the engine is running) and is equipped with zero-zero ejection seats (which enable ejections at low speeds and altitudes, even during ground mishaps).

It has a maximum speed of 450 kmph, a range of 1,000 km, and can fly at an altitude of 6,500 metres (over 21,000 feet). It can also be equipped with rockets and bombs. HAL chose the Honeywell Garrett TPE331-12B turboprop engine to power the trainer. An agreement worth $100 million was signed on July 27, 2022 between HAL and Honeywell for 88 engines, kits, maintenance and support services.

Being an indigenous design, the turboprop trainer is designed to configure upgrades to incorporate future requirements of the armed forces. The HTT-40 has nearly 56 per cent indigenous content, which will progressively increase to over 60 per cent through the indigenisation of components and subsystems. According to the MoD, the HTT-40 programme is supported by a supply

HOMEMADE AEROBATICS

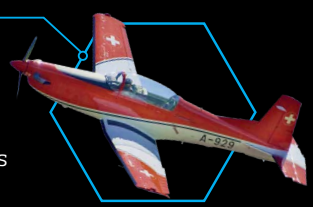
- HTT-40 is a twin-seater, fully aerobatic turboprop aircraft with modern avionics
- Hot-refuelling capability, zero-zero ejection seats
- Maximum speed of 450 kmph, range of 1,000 km, can fly at altitude of 6,500 metres
- It has 56 per cent indigenous content, will progressively increase to over 60 %.
- Cost: about $7 million per aircraft

IAF'S TRAINER AIRCRAFT

IAF operates 260 trainer aircraft against its requirement for 388

C-7 Mk II

Basic trainer jet, manufactured by Swiss firm Pilatus; Warranty of PC-7 fleet with Pilatus has expired, forcing IAF to source PC-7 spares from the open market. Numbers in service: **75**





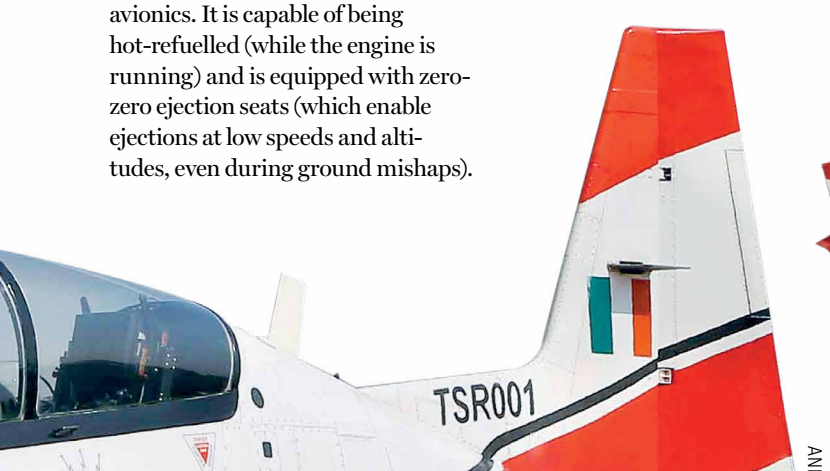
Hawk Mk 132

Advanced jet trainer, manufactured by BAE Systems of UK origin; licensed production in India by HAL. Numbers in service: **102**



HJT-16 Kiran

Intermediate jet trainer, manufactured by HAL; Numbers in service: **82**



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HTT-40

Manufactured by HAL; number commissioned by Centre: **70**, to be delivered over 6 years. First two to be delivered within 20 months